

- Finding that our dataset is very imbalanced (<10% is sepsis positive)
 - Using **XGboosted boosted tree model** to help with the pos_weight
 - AND used SMOTE to synthetically create more sepsis positive cases by looking at distances in between two sepsis cases and choosing something in between
 - 129 to 1790 sepsis positive samples

Suggested model parameters for 900+ parameters (both structured + unstructured inputs)

```
# 2. Re-train XGBoost on the balanced data
# Note: We remove scale_pos_weight because the data is now balanced 1:1
xgb_balanced = xgb.XGBClassifier(
    objective='binary:logistic',
    n_estimators=300,          # Increased, but we'll use a lower learning rate
    learning_rate=0.02,       # Slow down learning to avoid memorizing noise
    max_depth=4,              # Keep trees shallow to prevent overfitting 970 features

    # --- NEW REGULARIZATION PARAMETERS ---
    reg_alpha=1.0,            # L1 Regularization (forces unimportant feature weights to 0)
    reg_lambda=2.0,           # L2 Regularization (prevents any one feature from dominating)
    colsample_bytree=0.3,     # Only look at 30% of features for each tree (CRITICAL for 970 features)
    subsample=0.8,            # Use 80% of rows for each tree

    eval_metric='aucpr',
    random_state=42
)
```

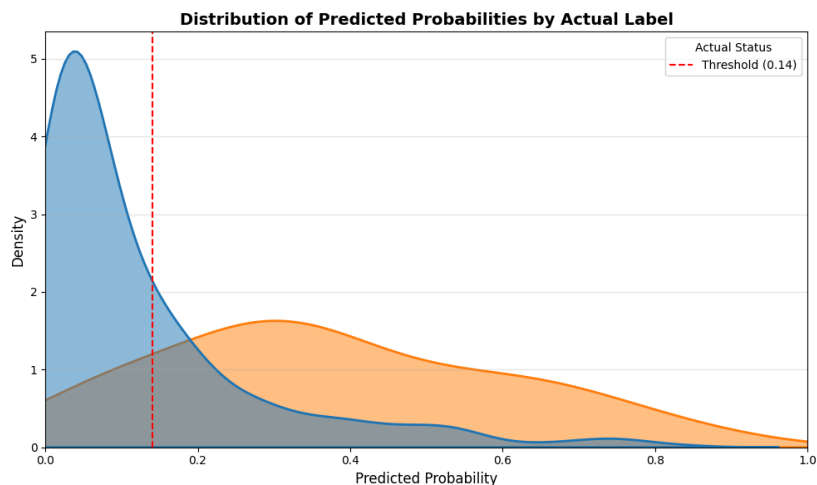
- Results using their default param values

```
... --- Results (Decision Threshold: 0.14) ---
              precision    recall  f1-score   support

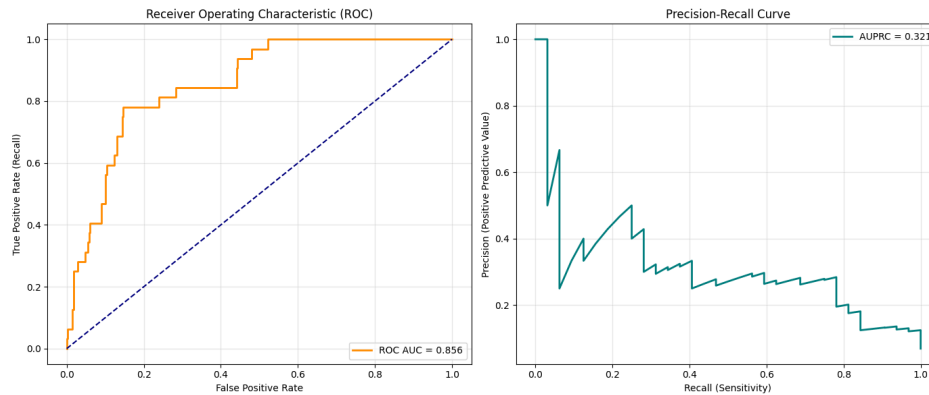
   No Sepsis       0.98      0.72      0.83       431
    Sepsis         0.18      0.84      0.30        32

 accuracy          0.58
 macro avg         0.58      0.78      0.56
 weighted avg      0.93      0.73      0.79

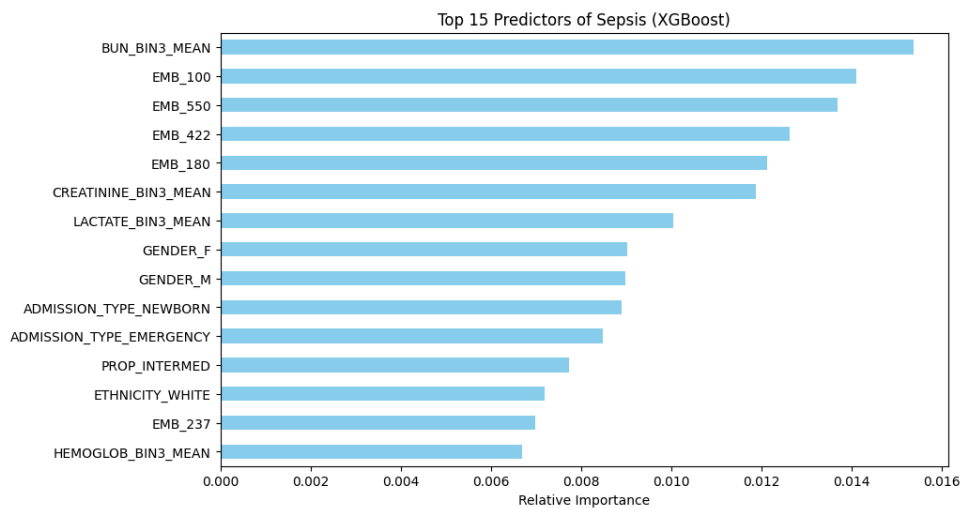
F2-Score (Recall-focused): 0.4874
```



* blue is no sepsis, orange is sepsis



Current contributing features after removing possible post-sepsis parameters:



- Took out NUM_ANTIBIOTICS column because it was the top predictor, but it probably occurs after a sepsis diagnosis IRL
- Shifted threshold from 0.5 (default) to 0.14 based on some algorithm to get 80% recall for sepsis positive
 - Meaning 80% of true positives is identified as true positive
- Random idea - if not for clinical use, this application can help create more balanced datasets by saving the positive flags (regardless of if it's actually positive, because its accuracy rate will still allow a more balanced dataset)

RESULTS when using just structured data

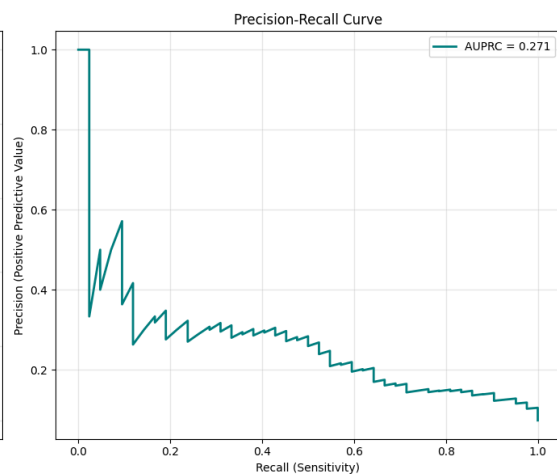
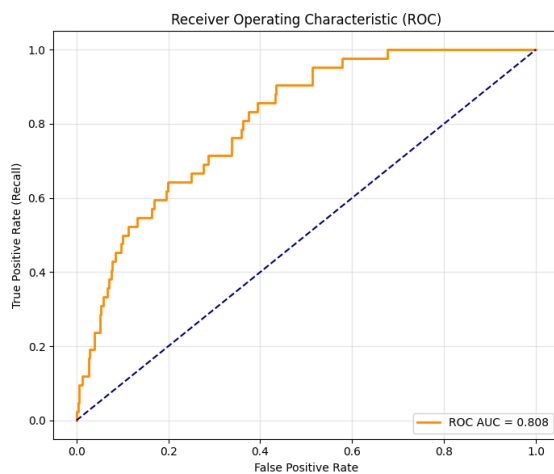
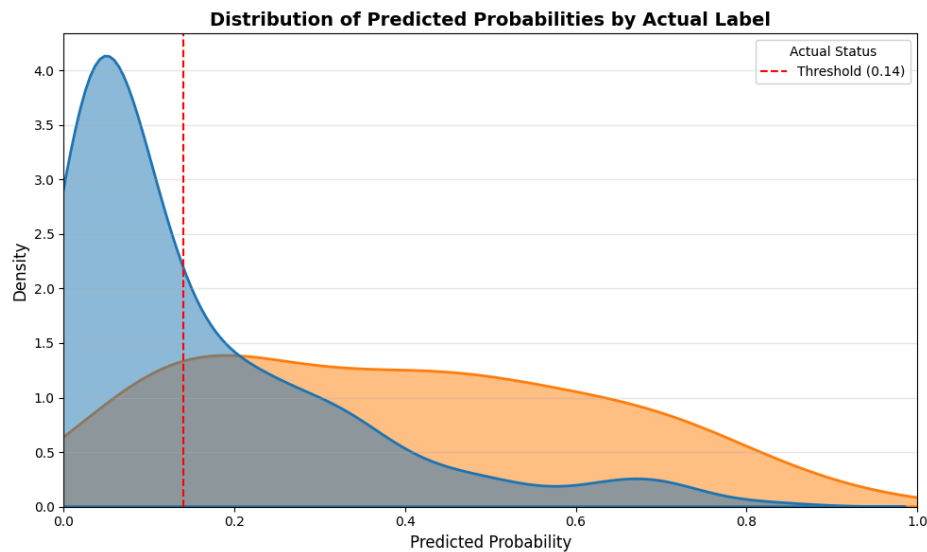
- Performance varied a little between the different time bins

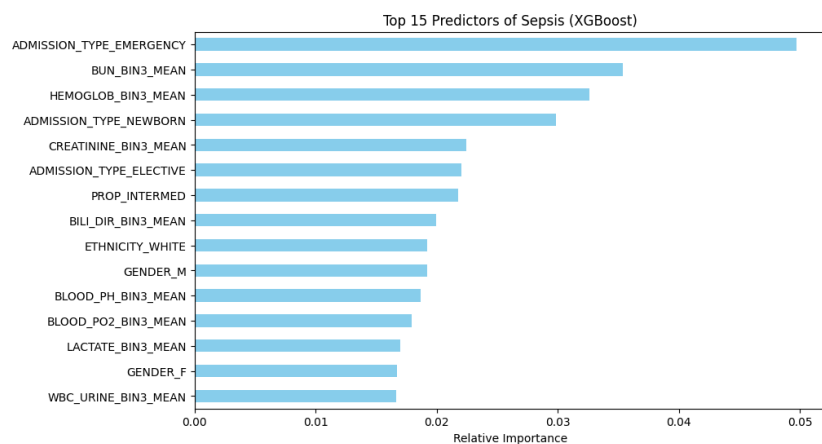
I..e 18 hours

```
... --- Results (Decision Threshold: 0.14) ---
```

	precision	recall	f1-score	support
No Sepsis	0.98	0.63	0.77	529
Sepsis	0.15	0.81	0.25	42
accuracy			0.64	571
macro avg	0.56	0.72	0.51	571
weighted avg	0.92	0.64	0.73	571

F2-Score (Recall-focused): 0.4282





18 hrs:

```

... --- Results (Decision Threshold: 0.14) ---
      precision    recall  f1-score   support

   No Sepsis      0.98      0.63      0.77      529
    Sepsis       0.15      0.81      0.25       42

   accuracy              0.64      571
  macro avg              0.56      0.72      0.51      571
 weighted avg              0.92      0.64      0.73      571

F2-Score (Recall-focused): 0.4282

```

15 hrs:

```

--- Results (Decision Threshold: 0.14) ---
      precision    recall  f1-score   support

   No Sepsis      0.98      0.61      0.75      529
    Sepsis       0.15      0.88      0.26       42

   accuracy              0.63      571
  macro avg              0.57      0.74      0.51      571
 weighted avg              0.92      0.63      0.72      571

F2-Score (Recall-focused): 0.4490

```

12 hrs:

```

... --- Results (Decision Threshold: 0.14) ---
      precision    recall  f1-score   support

   No Sepsis      0.98      0.63      0.76      529
    Sepsis       0.15      0.83      0.25       42

   accuracy              0.64      571
  macro avg              0.56      0.73      0.51      571
 weighted avg              0.92      0.64      0.73      571

F2-Score (Recall-focused): 0.4364

```

09 hrs:

```

... --- Results (Decision Threshold: 0.14) ---
      precision    recall  f1-score   support

   No Sepsis      0.98      0.63      0.77      529
    Sepsis       0.16      0.86      0.26       42

   accuracy              0.65      571
  macro avg              0.57      0.74      0.52      571
 weighted avg              0.92      0.65      0.73      571

F2-Score (Recall-focused): 0.4500

```